

Caution
Clockwise rotation of the scan frame may
cause damage to the slip ring brushes.
Ring brushes rotate the scan frame
counter clockwise only.

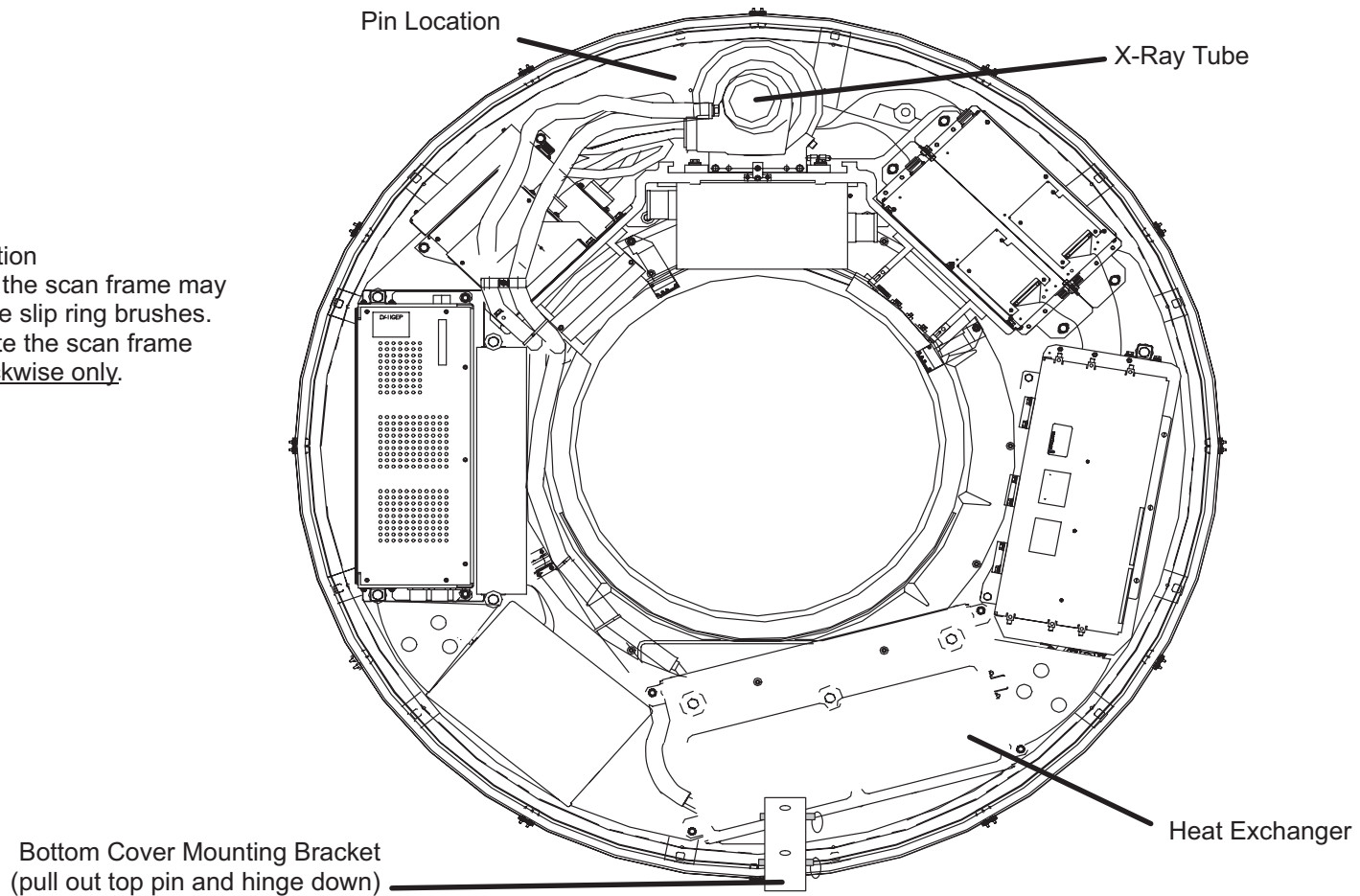


FIGURE 248 SCANNED FRAME PINNED WITH TUBE AT 12:00 POSITION

4. Secure Hoist Assembly to Patient Support. Make sure Clamp Base is seated firmly in position on Patient Support and that Knob Bolts are screwed in securely. See Figure 249.

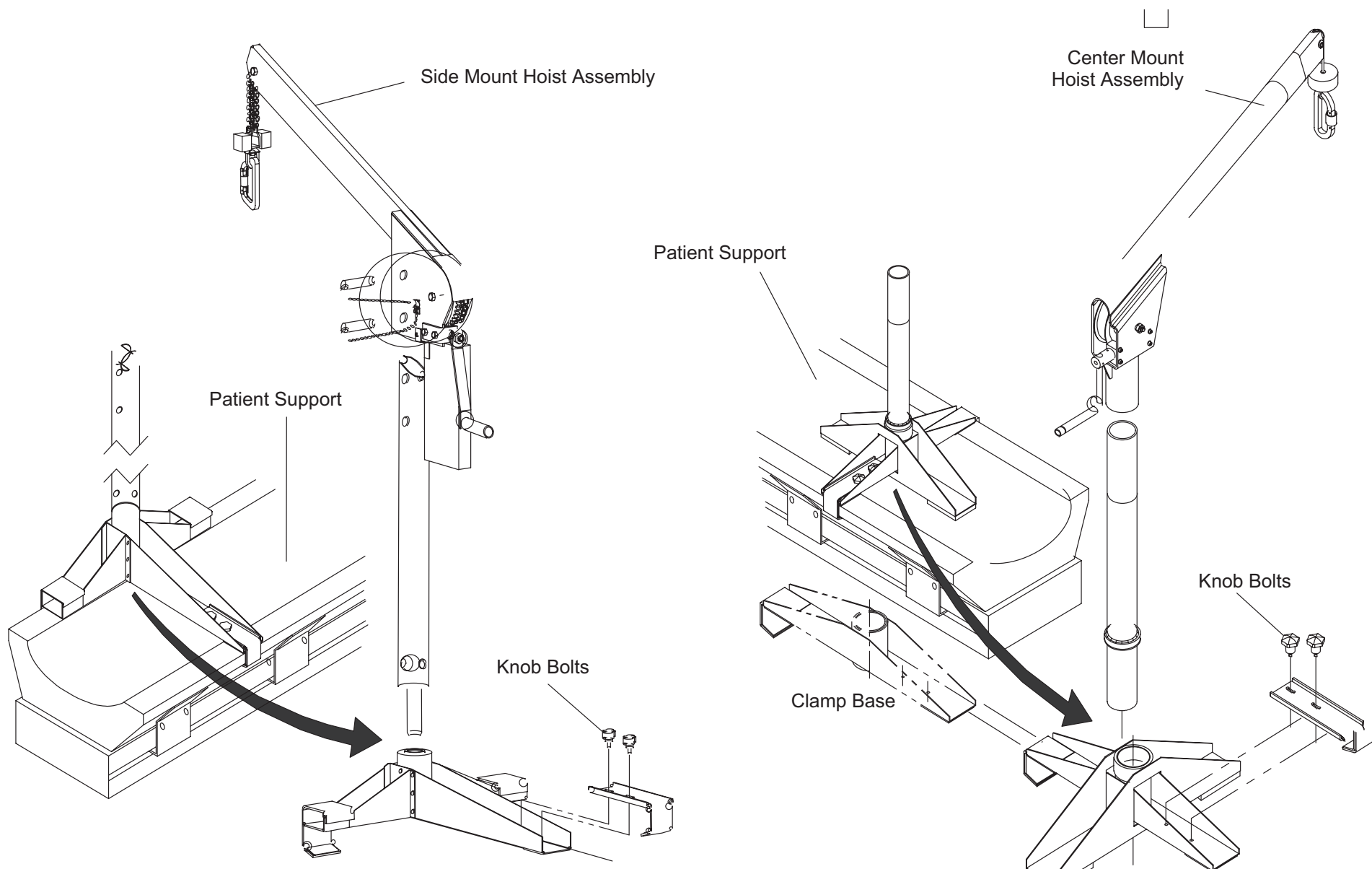


FIGURE 249 HOIST ASSEMBLY

5. Loosen and remove the four Cradle Locking Hex Head Bolts. See Figure 250.

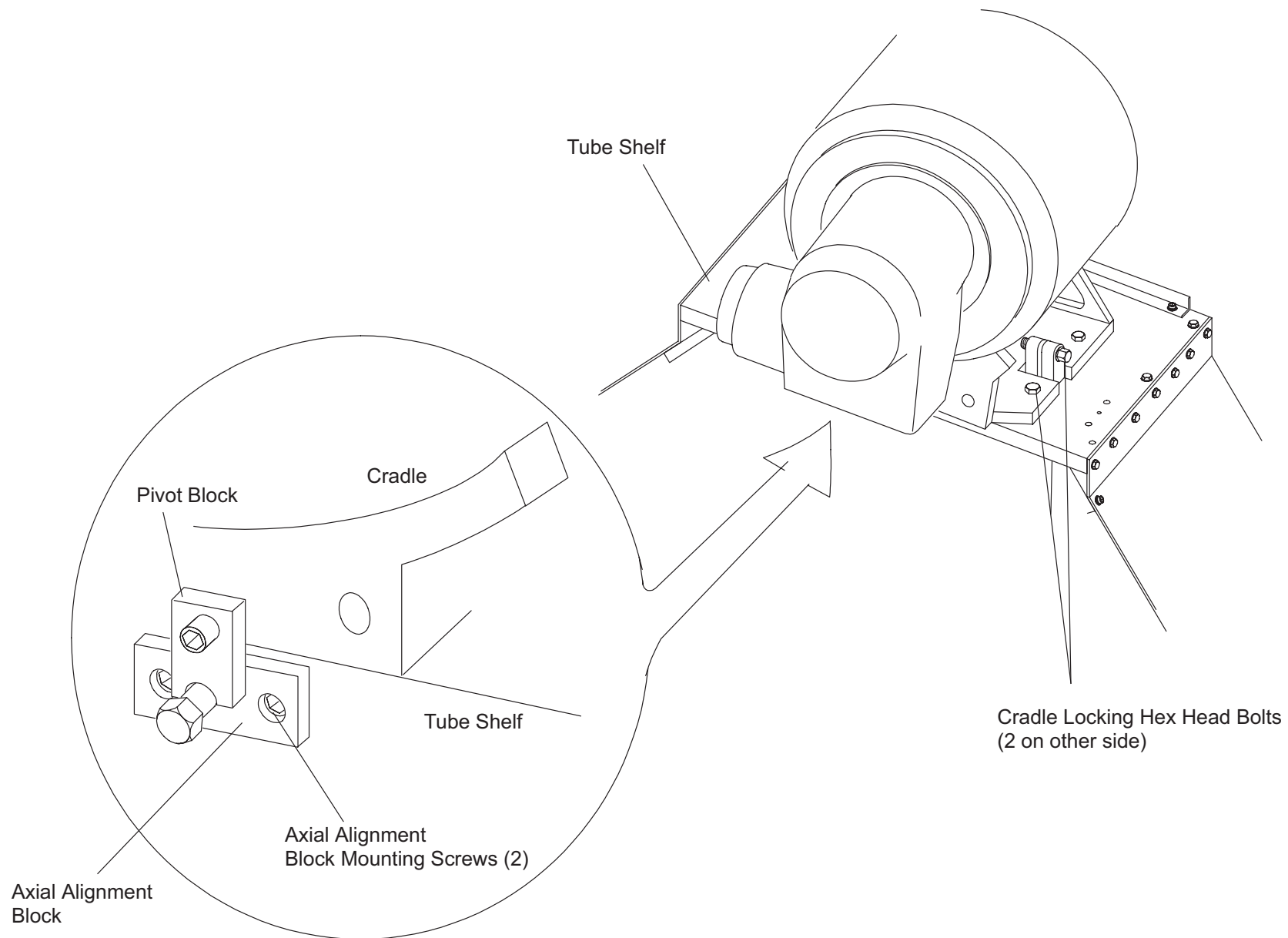


FIGURE 250 XRAY TUBE MOUNTING

6. Install the tube mounting tray in front of the xray tube and secure with the appropriate bolts. Refer to Figure 251.

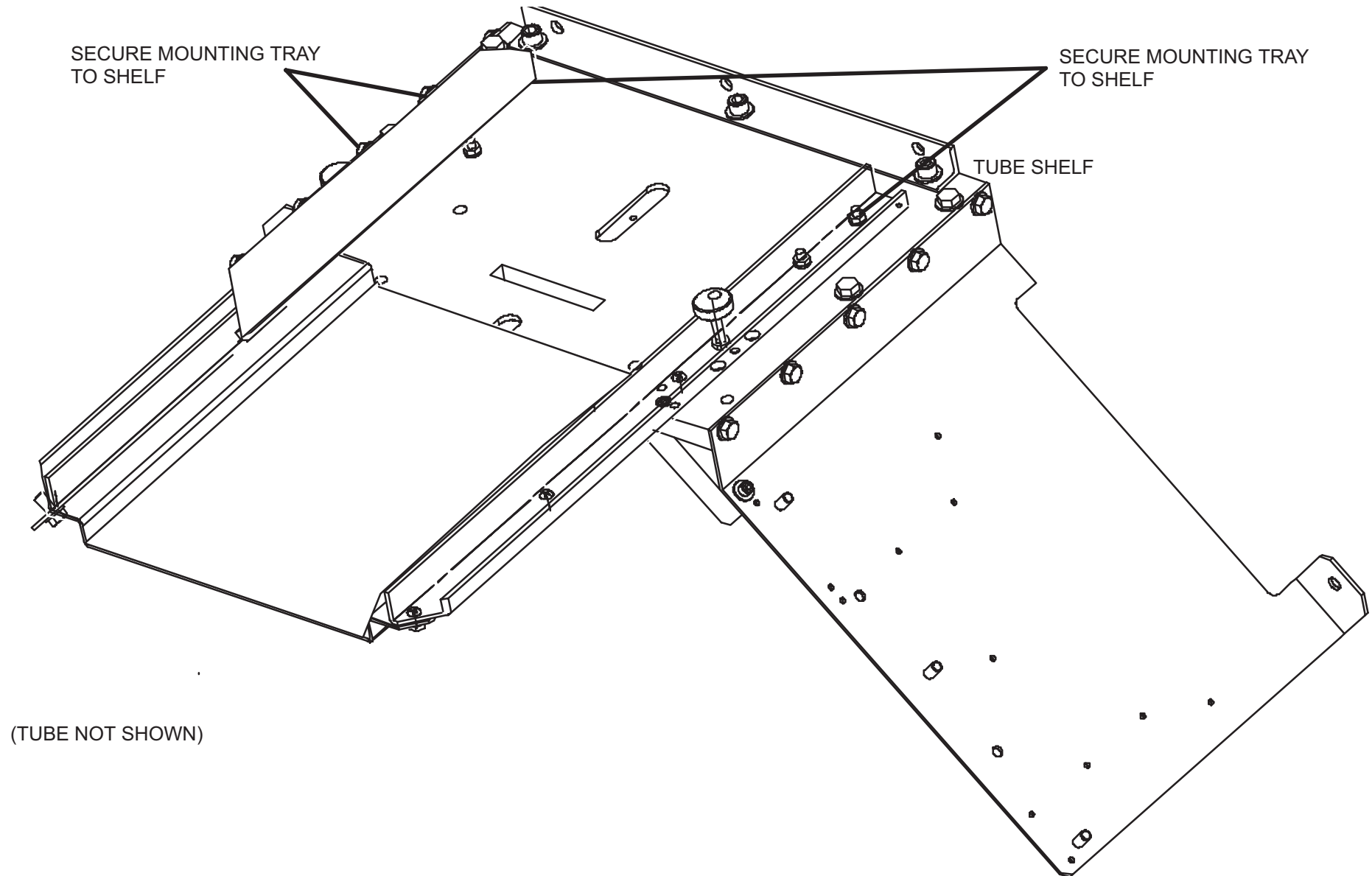


FIGURE 251 XRAY TUBE MOUNTING TRAY ON TUBE SHELF

7. Tag and disconnect the Quick Disconnects labeled J4 (Rotor Cable) and J6 (Thermal Switch Cable) that are located just behind and to the left of the X-Ray Tube.
8. Tag and remove High Voltage Federal Connectors from Anode and Cathode High Voltage Modules. Touch each High Voltage Federal Connector conductor to ground to bleed off any residual capacitive charge.



WARNING

MAKE NO ATTEMPT TO REMOVE THE HIGH VOLTAGE FEDERAL CONNECTORS FROM THE X-RAY TUBE. THEY ARE PERMANENTLY ATTACHED AND STAY WITH THE TUBE. REMOVING THESE CONNECTORS WILL CAUSE SERIOUS DAMAGE AND/OR DESTRUCTION OF THE TUBE.

9. Loosen one of the two Axial Alignment Block Mounting Screws and remove the other. Let the Block swing down to allow removal of the X-Ray Tube Cradle. Refer to Figure 250.

NOTE

This method minimizes the need for lateral adjustment of the X-Ray Tube if the Axial Pivot Block is not loosened. However, care must be exercised not to damage the Patient Support. Place a protective cover on the Carbon Top before setting the Tube and Cradle on it.

10. Remove Hose Clamps and any Tie Wraps securing Oil Hoses to Scan Frame. Pull X-Ray Tube out along mounting tray (6"–10" max) in order to route Lifting Strap through Cradle Holes. See Figure 252.

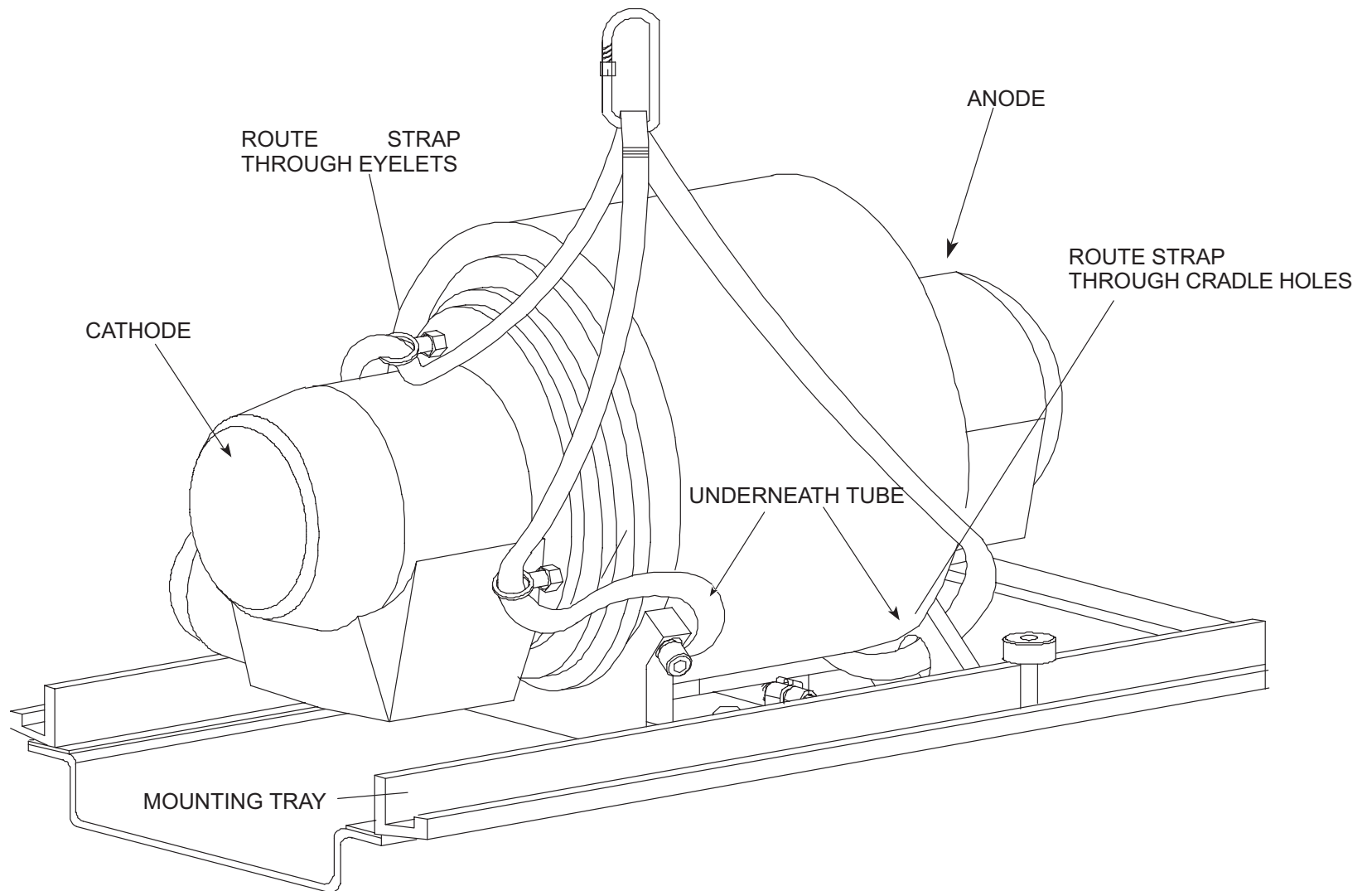


FIGURE 252 LIFTING STRAP ROUTING

11. Route Lifting Strap through one eyelet, under X-Ray Tube and up other side through other eyelet. Route second Lifting Strap through one Cradle hole at other end of X-Ray Tube, under X-Ray Tube and through hole on opposite side. Refer to Figure 252.

12. Place Lifting Strap ends onto the Hoist Clip and lock Clip. Remove slack and carefully pull Tube and Cradle Assembly to the end of Mounting Rails.
13. Carefully pull Tube Assembly from Mounting Rails. Brace Tube Assembly to keep it from rocking.
14. Carefully move Patient Support away from Gantry, swing X-Ray Tube to the right and lower Tube to floor. Lay Tube on its back with Shutter side up.
15. Remove Lifting Straps from Tube and Cradle Assembly and then remove Cradle from Tube.
16. Tag and disconnect the Quick Disconnect labeled J5 (Heat Exchanger Power) at the High Speed Starter.
17. Remove the two bolts on the heat exchanger supports. Slide the heat exchanger out and away from the gantry. See Figure 253.

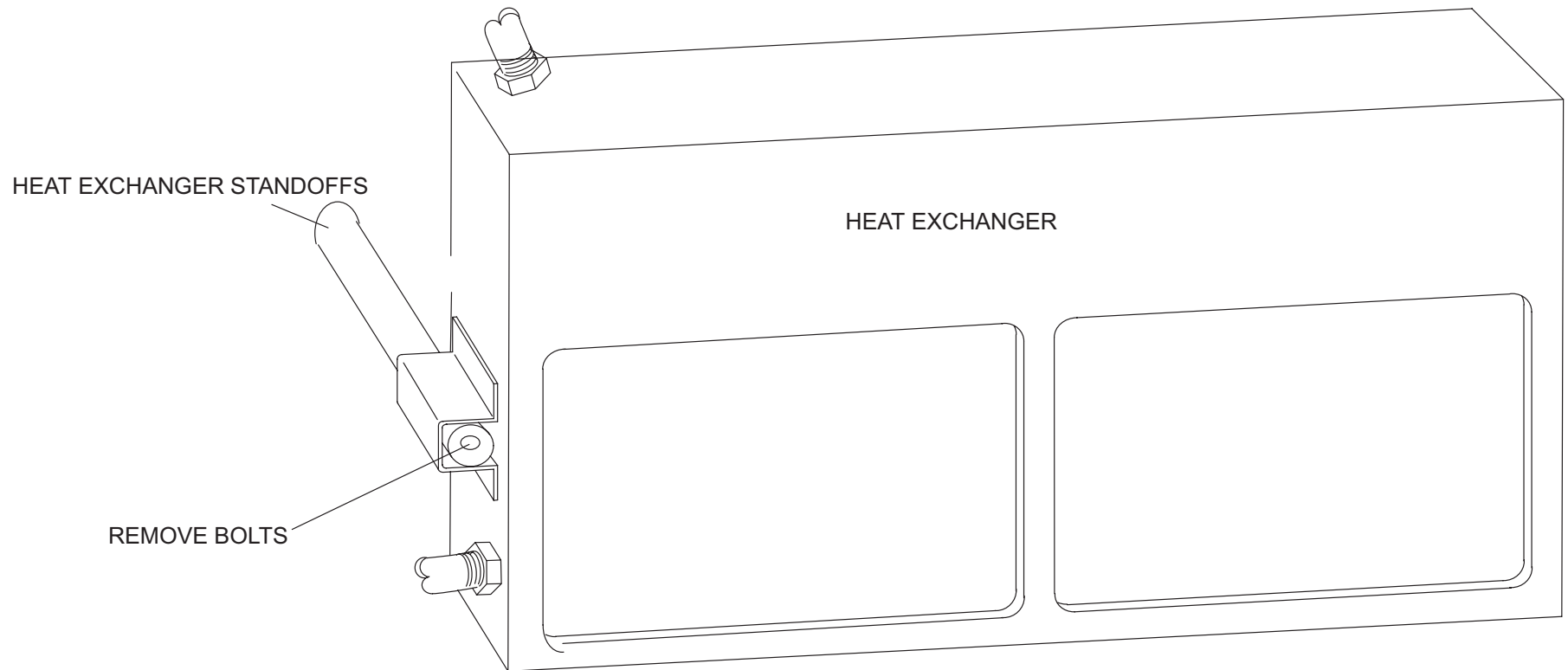


FIGURE 253 HEAT EXCHANGER MOUNTING HARDWARE



CAUTION

THE HEAT EXCHANGER WEIGHS 45 LBS. USE CAUTION WHEN HANDLING THIS ITEM. FAILURE TO COMPLY MAY RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY TO PERSONNEL.

18. Remove the Heat Exchanger Mounting bolts. Set them aside for later installation on the replacement unit.

XRAY TUBE REPLACEMENT:

19. Place box with replacement X-Ray Tube on left side of Patient Support. Open shipping container of replacement Tube and remove packaging material from container.
20. Lift Heat Exchanger from container and place it on floor near front of Gantry.

NOTE

Cover the Patient Support before placing the Tube on it or the Carbon Top may be damaged by the heavy Tube.

21. Place Lifting Straps through eyelets of Tube. Lift the X-Ray Tube from its vertical position within its container and position it with the Cradle Side up on the Patient Support.
22. With Tube Port facing upwards, remove protective metal covering. Check for the plastic covering over Tube Port opening and check for dirt/dust. Remove any debris found on the plastic. **DO NOT REMOVE PLASTIC**

23. Attach Cradle Assembly to Xray Tube. Follow illustration in Figure 254 for proper installation. Torque Mounting Hex Head and Socket Head Screws to 15 ft-lbs. alignment Pins.

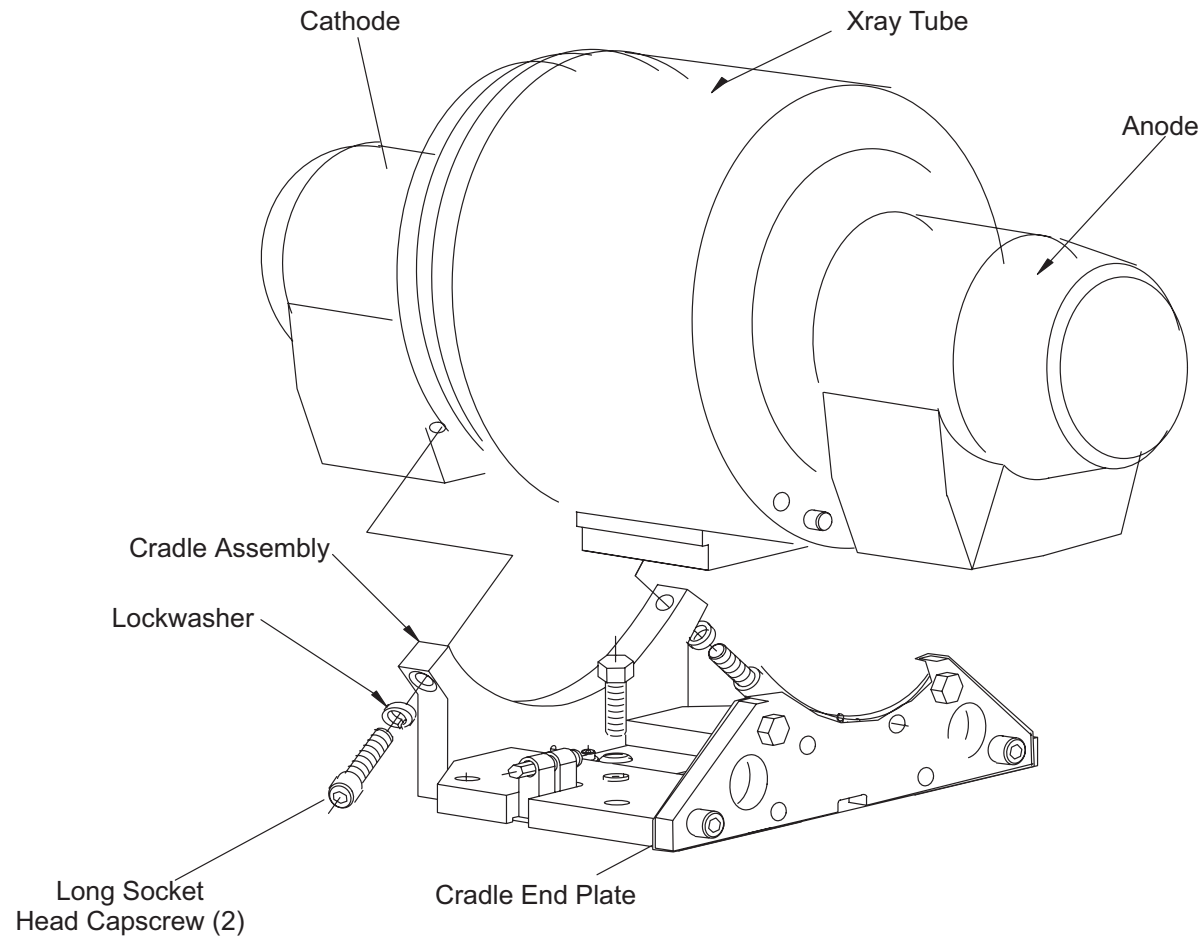


FIGURE 254 Cradle Assembly

24. Remove Packing Slips and Cable Ties from the Tube. Wipe off the Tube with cloth and check for container-debris.
25. (If possible) Rotate the Tube so that its Cradle side is facing down and its Oil Hoses facing left on the Patient Support.

NOTE

Ensure the Carbon Top has a protective cover to prevent damage by the Axial Alignment Pivot Block attached to the Cradle.

26. Remove the Lifting Straps from the eyelets. Reposition one strap through one eyelet, under X-Ray Tube and up other side through other eyelet. Route second strap through one cradle hole, under X-Ray Tube and through hole on opposite side. Refer to Figure 252.
27. Hoist the Tube to the tube tray. Push tube onto tray until back of cradle assembly is flush with front of Tube Shelf.



WARNING

USE EXTREME CAUTION AND VERIFY THAT THE X-RAY TUBE IS FULLY SUPPORTED BY THE MOUNTING RAILS BEFORE REMOVING THE LIFTING STRAPS. FAILURE TO COMPLY MAY RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY TO PERSONNEL.

28. Remove Lifting Straps from around Tube and move the hoist assembly out of the way.
29. Check Cables for clearance around the Tube. Push the Tube all the way back onto the Tube Shelf, ensuring the Cradle Assembly is seated on the Key Block. Refer to Figure 252.
30. Replace the Cradle Locking Bolts and fasten the Cradle down enough to relieve pressure from the tube tray.
31. Remove tube tray and secure Axial Alignment Block in its original position with the Socket Head Capscrew removed earlier.
 - Apply Loc-Tite to Alignment Block Screws and tighten to hand-tight.
 - Tighten Cradle hardware to 15 ft-lbs.
32. Position the Heat Exchanger on standoffs and push it back into position. Place the Cables behind the Heat Exchanger. Secure with the two bolts. Refer to Figure 253.
33. Secure Oil Hoses with Hose Clamps. DO NOT OVERTIGHTEN HEX NUTS.

NOTE

Align the Hose Clamps with the labels on the Hoses.

34. Reconnect Quick Disconnects J4 (Rotor Cable), J5 (Heat Exchanger Power Cable), and J6 (Thermal Switch Cable).
35. Inspect the Federal Connector for any sign of damage.
36. Using a clean, lint-free wipe and 90% alcohol, wipe the surface clean of dirt or dust.
37. Pour enough Transformer Oil into the Federal Connector receptacles to coat the entire surface of the cable connector after insertion into the module (10–15 cc). The amount will vary due to slight differences in the connectors.
38. Replace the Anode and Cathode High Voltage Federal Connectors.
39. Rotate the Scan Frame counterclockwise by hand while checking for loose Cables. Also check for clearance over the entire range of travel between the Rotating Frame Cables and the Stationary Frame.



CAUTION

CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.

40. Align the X-Ray Tube. Refer to [X-Ray Tube Mechanical Alignment](#) in the Mechanical Calibration Manual.
41. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.



WARNING

USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.

42. Push the bottom cover mounting bracket back up and secure with the locking pin removed earlier. Replace the Gantry Front Cone. Refer to Front Cone Replacement in the Gantry Covers procedure.

NOTE

Step 42. is necessary to avoid artifacts in the test scans.

43. Pull the Front Cover Interlock Switch Plunger out to enable operation while the Front Covers are open.
44. To verify proper System operation, run Full-field Water Phantom Scans and allow the System to process and display an Image.
45. Return the Front Cover Interlock Switch Plunger to its center position.
46. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
47. Replace the Gantry lower Front Cover, close the upper Front Cover. Refer to Upper Front Gantry Covers Closing procedure in the Gantry Covers procedure.